

A clinical prognostic scoring system for Guillain-Barré syndrome



Rinske van Koningsveld, Ewout W Steyerberg, Richard A C Hughes, Anthony V Swan, Pieter A van Doorn, Bart C Jacobs

Summary

Background Guillain-Barré syndrome (GBS) is an acute post-infectious immune-mediated peripheral neuropathy with a highly variable clinical course and outcome. We aimed to develop and validate a scoring system based on clinical characteristics in the acute phase of GBS to predict outcome at 6 months.

Methods We studied patients with GBS who were unable to walk independently. A derivation set included 388 patients from two randomised controlled trials and one pilot study. Potential predictors were assessed for their association with the inability to walk independently at 6 months. A simple clinical scoring system was developed on the basis of regression coefficients of predictors in a multivariable logistic regression model. Model performance was quantified with respect to discrimination (area under receiver operating characteristics curve, AUC) and calibration (graphically). We validated our scoring system in a set of 374 patients from another randomised trial.

Findings We included three variables that were predictive of poor outcome at 6 months in our model: age, preceding diarrhoea, and GBS disability score at 2 weeks after entry. Scores ranged from 1 to 7, with three categories for age, two for diarrhoea, and five for GBS disability score at 2 weeks. Predictions corresponding to these prognostic scores ranged from 1% to 83% for the inability to walk independently at 6 months. Predictions agreed well with observed outcome frequencies (adequate calibration) and showed a very good discriminative ability (AUC 0.85) in both data sets.

Interpretation A simple scoring system for patients with GBS, based on three clinical characteristics, accurately predicts outcome at 6 months. The system could be used to counsel individual patients and identify high-risk groups to guide future trials.

Introduction

Guillain-Barré syndrome (GBS) is a post-infectious immune-mediated peripheral neuropathy characterised by rapidly progressive weakness and sensory loss, usually followed by slow clinical recovery. GBS is heterogeneous in severity of neurological deficits and prognosis. Some patients develop paralysis of oculomotor, facial, oropharyngeal, respiratory, and limb muscles within days and remain bedridden or wheelchair bound. Others have mild limb paresis from which they recover spontaneously within weeks. Advances in general care facilities and the availability of specific treatments have improved the outcome of patients with GBS. Nonetheless, about 20% of patients die from the complications of GBS or remain severely disabled.^{1,2} To improve outcome in this subgroup, future trials should focus on patients with poor prognosis, identified in the early phase of disease in which additional treatment might still be effective.³ Outcome from GBS is presumably determined by the extent of nerve damage in the acute phase and the capacity to recover in the convalescent phase. Previous studies showed that preceding infection, age, rapid progression, disability at nadir, and electrophysiological characteristics were associated with long-term prognosis.⁴⁻⁹ However, readily applicable and validated models are not yet available to predict outcome in the early phase of the disease.

We aimed to develop a simple yet accurate prognostic model for GBS on the basis of clinical characteristics that are easily obtained in the acute phase of disease. Two large cohorts of patients with GBS were used to develop and validate such a model to predict outcome at 6 months.

Methods

Participants

All patients fulfilled the diagnostic criteria for GBS.¹⁰ The derivation set contained 397 patients with GBS enrolled in two randomised controlled trials and one pilot study.¹¹⁻¹³ Most patients were randomised in Dutch centres; the others in two Belgian centres and two German centres. In the first study, 147 patients were included between 1985 and 1991 in a multicentre double-blind randomised controlled trial that compared the effect of plasma exchange with intravenous immunoglobulin.¹¹ In a pilot study in the Netherlands, 25 patients were included to investigate the additional effect of methylprednisolone on standard treatment with intravenous immunoglobulin.¹² In a third study, this combination was tested against intravenous immunoglobulin and placebo in a multicentre double-blind randomised controlled trial done between 1994 and 2000 and including 225 patients.¹³ Similar inclusion and exclusion criteria were used in these three studies. Inclusion criteria were fulfilment of the diagnostic

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Departments of Neurology

(R van Koningsveld MD,

Prof P A van Doorn MD,

B C Jacobs MD), Public Health

(Prof E W Steyerberg PhD), and

Immunology (B C Jacobs),

Erasmus MC, Rotterdam,

Netherlands; and Department

of Clinical Neuroscience, King's

College London, Guy's Hospital,

London, UK

(Prof R A C Hughes MD,

A V Swan PhD)

Correspondence to:

Dr Bart C Jacobs, Department of

Neurology, Erasmus MC, PO Box

2040, 3000 CA Rotterdam,

Netherlands

b.jacobs@erasmusmc.nl

criteria for GBS, onset of weakness within 2 weeks before study entry, and being unable to walk 10 m across an open space without assistance (GBS disability score at least 3). Exclusion criteria were age below 6 years, previous GBS, severe allergic reaction to properly matched blood products, known selective IgA deficiency, pregnancy, previous steroid treatment, or contraindications for steroid treatment (not in the plasma exchange/intravenous immunoglobulin trial),

having severe concurrent disease, or inability to attend follow-up.

Our validation set consisted of 379 patients enrolled in a multicentre double-blind randomised controlled trial.¹⁴ Patients were included between 1993 and 1995 in 38 centres in 11 countries in Europe, North America, and Australia. In this trial the effect of plasma exchange was compared with intravenous immunoglobulin and with the combination of plasma exchange and intravenous immunoglobulin. Inclusion criteria were the same as those in the trials described above except that the lower age limit was 16 years. Patients with atypical forms of GBS, serious pre-existing disease, or contraindications to plasma exchange or intravenous immunoglobulin were excluded.

Baseline characteristics and outcomes from all four studies have been described in detail before,^{11–14} and were much the same between centres and seasons, and between patients enrolled in different years.

Data collection

All demographic, clinical, and laboratory data were collected prospectively. In the Dutch trials cranial nerve dysfunction, GBS disability score, and Medical Research Council (MRC) sum score were measured every week during the first 8 weeks after entry, every 2 weeks during weeks 9–14, and every 4 weeks during weeks 15–26 until the patient reached the functional score of one or less. Follow-up ended after 26 weeks.

The GBS disability score and the MRC sum score both give an indication of the severity of disease. The MRC sum score was defined as the sum of MRC scores from six muscles in the upper and lower limbs on both sides so that the score ranged from 60 (normal) to 0 (quadriplegic; panel). Sensory deficits (of any modality) were scored at entry and at 4, 8, 14, and 26 weeks after randomisation. The GBS disability score is a widely accepted scoring system to assess the functional status of patients with GBS in which scores range from 0 (normal) to 6 (dead; panel). We defined poor outcome as a GBS disability score at 6 months of 3 or more, which corresponds with the inability to walk 10 m independently. Fairly good outcome was defined as a GBS disability score at 6 months of 2 or less. The endpoint was determined at 6 months after inclusion because most of the recovery process will have occurred by this time in those patients who do recover.

Other tests done included routine testing of blood, urine, and cerebrospinal fluid; serological screening for preceding infections with *Campylobacter jejuni*, cytomegalovirus, Epstein-Barr virus, and *Mycoplasma pneumoniae*; and testing for IgM and IgG antibodies to the gangliosides GM1 and GD1a in pretreatment serum samples. The sensitivity and specificity of the infection serology in GBS is unknown because of the delay between the preceding infection and hospital admission. ELISA is currently the most appropriate standard technique to demonstrate the presence of serum anti-ganglioside antibodies.¹⁷

	Number of patients	Number of patients with poor outcome	Odds ratio (95% CI)	p value
Demographic features				
Total	388	71 (18%)		
Age (years)				
≤40	137	13 (9%)	1	0.001
40–60	111	19 (17%)	2.0 (0.9–4.2)	
>60	140	39 (28%)	3.7 (1.9–7.3)	
Sex (male)	213	41 (19%)	1.6 (0.7–1.9)	0.6
Infections and serology				
Symptoms preceding infection*				
Diarrhoea	88	27 (31%)	2.6 (1.5–4.5)	0.001
Upper respiratory tract infection	143	17 (12%)	0.5 (0.3–0.9)	0.01
Infection serology				
<i>Campylobacter jejuni</i>	105	30 (29%)	2.7 (1.5–4.7)	0.001
Cytomegalovirus	42	7 (17%)	0.9 (0.4–2.2)	0.9
Epstein-Barr virus	41	9 (22%)	1.4 (0.6–3.1)	0.4
<i>Mycoplasma pneumoniae</i>	9	1 (5%)	0.3 (0.03–1.9)	0.2
Anti-ganglioside antibodies				
GM1	74	18 (24%)	1.7 (0.9–3.2)	0.1
GD1a	20	4 (20%)	1.2 (0.4–3.7)	0.8
Neurological deficits				
GBS disability score at entry				
3	91	7 (8%)	1	0.003
4	261	52 (20%)	3.0 (1.3–6.8)	
5	36	12 (33%)	6.0 (2.1–17)	
GBS disability score at 2 weeks				
0 or 1	15	0 (0%)	0	<0.0001
2	84	2 (2%)	1	
3	65	2 (3%)	1.3 (0.2–9.5)	
4	144	30 (21%)	11 (2.5–46)	
5	80	37 (46%)	35 (8.1–153)	
MRC sum score at entry				
60–51	47	0 (0%)	0	<0.0001
50–41	179	20 (11%)	1	
40–31	79	22 (28%)	3.1 (1.6–6.0)	
30–21	50	17 (34%)	4.1 (1.9–8.6)	
20–0	32	12 (38%)	4.8 (2.0–11)	
Weakness before entry <4 days	191	49 (26%)	2.7 (1.6–4.7)	<0.0001
Cranial nerve involvement	215	40 (19%)	1.1 (0.6–1.8)	0.7
Sensory deficits	315	61 (19%)	1.5 (0.7–3.1)	0.3

MRC=Medical Research Council. *Symptoms of an infection in the 4 weeks preceding the onset of weakness.

Table 1: Characteristics of the derivation set of 388 patients with GBS in relation to poor outcome at 6 months (inability to walk independently)

Panel: Clinical scores for GBS

GBS disability score¹⁵

- 0 A healthy state
- 1 Minor symptoms and capable of running
- 2 Able to walk 10 m or more without assistance but unable to run
- 3 Able to walk 10 m across an open space with help
- 4 Bedridden or chairbound
- 5 Requiring assisted ventilation for at least part of the day
- 6 Dead

Medical Research Council (MRC) sum score¹⁶

Sum of MRC scores of six muscle groups, including shoulder abductors, elbow flexors, wrist extensors, hip flexors, knee extensors, and foot dorsiflexors on both sides, ranging from 60 (normal) to 0 (quadriplegic). The MRC score of an individual muscle group ranges from 0 to 5:

- 0 No visible contraction
- 1 Visible contraction without movement of the limb
- 2 Active movement of the limb, but not against gravity
- 3 Active movement against gravity over (almost) the full range
- 4 Active movement against gravity and resistance
- 5 Normal power

In the validation set, the GBS disability score was assessed at randomisation and after 2, 4, 8, 12, 26, and 52 weeks. The presence or absence of sensory deficits during the neurological examination at randomisation was recorded.

Statistical analysis

Previous reports of randomised controlled trials and large case studies identified several acute phase clinical characteristics that were significantly associated with outcome in GBS.^{4–9} These potential prognostic factors were tested in the derivation set to determine their association with outcome. Factors were classified as demographic factors, preceding infections, or neurological deficits. Factors significantly associated with outcome in univariable analysis were further tested in multivariable analysis. A backward stepwise selection procedure was done with Akaike's information criterion.^{18,19} If two variables from the same category were more or less equally associated with outcome, we selected the variable most easily obtainable in clinical practice. We compared the prognostic values of the GBS disability score and MRC sum score at study entry and at 1 and 2 weeks after study entry. Since predictors had no more than 1% missing values, we imputed the missing values with the mean of the available values. We checked for linearity of continuous variables with restricted cubic splines.¹⁸

Model performance was quantified with respect to discrimination (area under receiver operating characteristics curve, AUC). The area ranges from

0·5 to 1·0 for sensible models.¹⁸ Calibration of predictions was assessed graphically by plotting observed frequencies against predicted probabilities. The internal validity of the regression model was assessed by bootstrapping techniques, which included the stepwise selection of prognostic factors for the model. This technique gives an impression of how over-optimistic the model is—ie, how much the performance of the model deteriorates when applied to a new group of similar patients.²⁰ The model was applied to the validation dataset for external validation. First, we used the regression coefficients as estimated in the derivation dataset. Next, the coefficients were recalculated in the validation dataset for comparison with estimates in the derivation dataset. A final scoring system was constructed based on the regression coefficients of the multivariable model in a data set where the derivation and validation sets were combined for larger reliability. We used the mean SE of the logit of the predicted probabilities per score to calculate CI.

Statistical analyses were done with Stata version 8.0, SPSS for Windows, and S+ software version 6.0.

Role of the funding source

The funding source had no role in study design, data collection, data analysis, data interpretation, or writing of the report. RvK, EWS, and BCJ had full access to all the data in the study. All authors had final responsibility for the decision to submit for publication.

Results

No primary endpoint was available for 12 (3%) of 397 patients from the derivation set, although in seven of these 12 patients the endpoint could be deduced by the principle of last observation carried forward. Four patients died within 2 weeks from study entry and were excluded, leaving 388 patients for model development. From the validation set of 379 patients, no primary endpoint was available in 11 (3%) patients, but in eight of these 11 patients, the endpoint could be extrapolated. Two patients died within 2 weeks from study entry and were excluded, leaving 374 patients for model validation.

Of the 388 patients in the derivation set, 71 (18%) were unable to walk independently at 6 months. Of these patients with poor outcome, nine died and 28 had a GBS disability score at 6 months of 3, 31 had a score of 4, and three had a score of 5. The mean age was 49 (SD 19) years and 213 (55%) were men. Of the 374 patients in the validation set, 83 (22%) individuals were unable to walk

	Derivation set (n=388)	Validation set (n=374)	Combined sets (n=762)
Age (per age category)	1·9 (1·3–2·7)	2·5 (1·7–3·6)	2·1 (1·6–2·8)
Preceding diarrhoea	3·7 (1·9–7·3)	4·6 (2·3–9·2)	4·1 (2·6–6·7)
GBS disability score at 2 weeks after entry (per point)	4·2 (2·7–6·5)	4·9 (3·1–7·8)	4·5 (3·3–6·2)

Table 2: Multivariable odds ratios (95% CI) from logistic regression models for the prediction of inability to walk at 6 months

	Categories	Score
Age at onset (years)	>60	1
	41–60	0.5
	≤40	0
Diarrhoea (≤4 weeks)	Absence	0
	Presence	1
GBS disability score (at 2 weeks after entry)	0 or 1	1
	2	2
	3	3
	4	4
	5	5
Erasmus GBS outcome score		1–7

Table 3: The Erasmus GBS outcome score

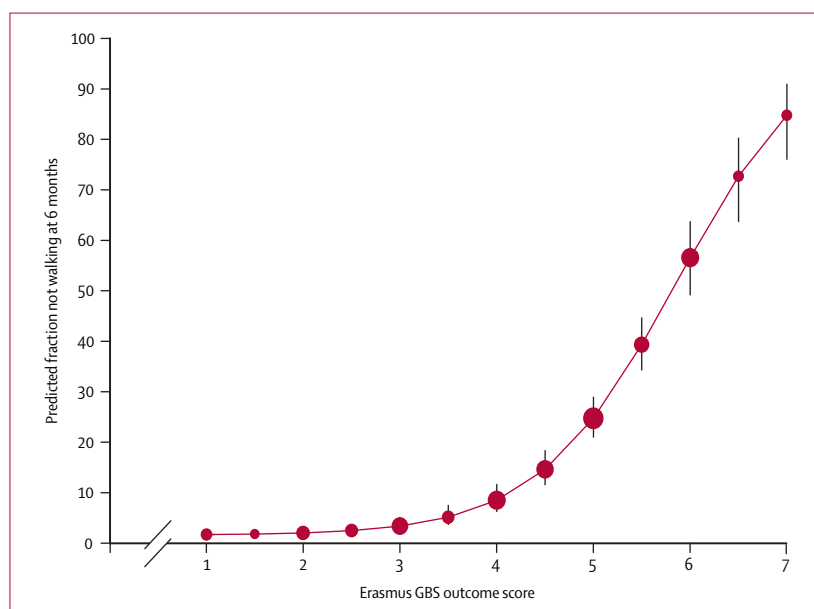


Figure: Predicted fraction of patients unable to walk independently at 6 months after randomisation on the basis of the Erasmus GBS outcome score (n=762)

Vertical bars indicate 95% CI. Point sizes proportionate to the number of patients with a specific score. The probability of not walking independently at 6 months is given by the equation $1/(1+\exp[8-2-1.4 \times \text{EGOS}])$.

	Derivation set (n=388)	Validation set (n=374)	Combined sets (n=762)
1–3	1/107 (1%)	0/86 (0%)	1/193 (0.5%, 0–2.9)
3.5–4.5	7/116 (6%)	9/110 (8%)	16/226 (7%, 4.1–11)
5	20/81 (25%)	23/80 (29%)	43/161 (27%, 20–34)
5.5–7	43/84 (51%)	51/98 (52%)	94/182 (52%, 44–59)
Total	71/388 (18%)	83/374 (22%)	154/762 (20%, 17–23)

Data are number unable to walk/number with particular Erasmus GBS outcome score (%) or number unable to walk/number with particular Erasmus GBS outcome score (%; 95% CI).

Table 4: Number of patients who were unable to walk independently at 6 months in the derivation and validation sets according to Erasmus GBS outcome score

independently at 6 months. The mean age of these patients was 52 (SD 17) years and 224 (60%) were men.

The results of the univariable analysis are shown in table 1. No association was found between outcome at 6 months and sex, cranial nerve involvement, sensory

deficits, serology for cytomegalovirus, Epstein-Barr virus, or *Mycoplasma pneumoniae* infection, antibodies to GM1 and GD1a, and the type of treatment. In multivariable analysis, age, preceding diarrhoea, and GBS disability score at 2 weeks after study entry emerged as the three main predictors of poor outcome at 6 months (table 2).

The AUC for a model that used age, preceding diarrhoea, and GBS disability score at 2 weeks after study entry was 0.85, indicating very good discriminative ability, which decreased by only 0.01 at internal validation. Use of GBS disability score at entry and at 1 week, and MRC sum score at entry, led to models with lower AUC (<0.80). Replacement of GBS disability score at 2 weeks with the MRC sum score at 1 or 2 weeks led to slightly better discriminating models (AUC>0.85). However, of the indicators of severity of disease, GBS disability score at 2 weeks was the only one that was also available in the validation set, and was therefore selected for use. Thus, the final model used age, preceding diarrhoea, and GBS disability score at 2 weeks after study entry as predictors of poor outcome at 6 months. The final model had similar discriminative ability in the external validation cohort of patients with GBS (AUC 0.85, 95% CI 0.81–0.89) as it did in the derivation set. The multivariable odds ratios were slightly larger in the validation set than in the derivation set (table 2).

We based the Erasmus GBS outcome score (EGOS) on the regression coefficients in the multivariable model of the combined datasets (n=762; table 2). Scores ranged from 1 to 7 with three categories for age, two for diarrhoea, and five for the GBS disability score at 2 weeks after study entry (table 3). As examples, we consider two bedridden patients with a GBS disability score of 4 (4 points). The first is 67 years old (1 point) and had a history of diarrhoea before GBS (1 point). The second is 37 years old (0 points) and had no history of diarrhoea (0 points). The Erasmus GBS outcome score of the first patient is thus 6, corresponding to a chance of not walking independently at 6 months of 55% (95% CI 48–62; figure). The Erasmus GBS outcome score for the second patient is 4, corresponding to a much better prognosis (only 7% [95% CI 5–10] chance of not walking independently at 6 months). For further illustration, patients were divided into four risk groups of roughly equal size (table 4). Only one (0.5%) of 193 patients with an Erasmus GBS outcome score of 3 or less was unable to walk independently at 6 months; by contrast, 94 (52%) of 182 patients with an Erasmus GBS outcome score of 5.5 or more were unable to walk independently at 6 months.

Discussion

The Erasmus GBS outcome score is a validated prognostic model that uses acute phase and easy-to-obtain clinical characteristics to determine outcome at 6 months in patients with GBS. On the basis of age, preceding

diarrhoea, and the GBS disability score at 2 weeks, the model accurately predicts the inability to walk independently at 6 months. This model provides a practical tool for physicians to inform individual patients about their prognosis, identify high-risk groups, and guide future treatment trials.

In six previous studies, older age and disease severity in the acute phase, expressed by various clinical parameters, were consistently identified as strong predictors of outcome.⁴⁻⁹ However, two of these studies partly overlap with our data.^{7,9} Preceding diarrhoea was a predictor of poor outcome in three studies.^{5,7,9} In two studies, preceding infections showed no relation with outcome, but the effect of diarrhoea was not specified.^{4,8} Here, preceding diarrhoea was a stronger prognostic factor than *Campylobacter jejuni* serology, which could be explained by the inaccuracy of the available serological tests to identify *C. jejuni* infections. Additionally, our study offers a model to predict outcome on the basis of an empirically weighted combination of predictors, based on two large cohorts with detailed and accurate information. Our model is similar in nature to prognostic models for other acute neurological diseases—eg, traumatic brain injury, stroke, and Bell's palsy—where outcome also depends on the extent of neural damage, as indicated by the severity of clinical symptoms in the acute phase, and the age-dependent capacity to recover.²¹⁻²³

In our analysis the GBS disability score at 2 weeks and the MRC sum score at 1 or 2 weeks had the highest predictive value for prognosis (AUC of at least 0.85). Which predictor is the best for describing disease severity in our model is debatable. The GBS disability score is probably the easiest to apply in clinical practice, although the MRC sum score might be more accurate. Moreover, a variable measured at 1 week supplies information in an early phase of the disease and therefore could be more helpful in early decisionmaking concerning therapeutic intervention. However, we could externally validate only the model based on the GBS disability score at 2 weeks. Nonetheless, we expect that the external validity would have been of similar quality for models with alternate indicators of disease severity.

Our study aimed to predict outcome in GBS by use of acute phase clinical features. Our model discriminates between the quarter of patients with less than 1% chance of disability and the quarter of patients with a 52% chance of disability (table 4), demonstrating the high predictive power of clinical features. Patients with the highest scores even had a chance of disability of up to 80%, although the number of patients with such scores was low (figure). Further improvement of prognostic models would require the inclusion of additional variables, although it is unlikely that these variables will be clinical characteristics. However, other studies have identified non-clinical features that are significantly related to outcome in GBS, including electrophysiological features of axonal

degeneration.^{4-6,8,9} These characteristics could help to refine the criteria for early identification of patients with poor prognosis, although their predictive power independent of the clinical features needs to be shown.

The Erasmus GBS outcome score was based on a distinct group of trial patients, which might restrict its general applicability. First, only severely affected patients who were unable to walk independently at study entry were included. The model has not been validated in less severe cases, although three prospective studies that included mildly affected patients reported older age and disease severity as prognostic factors associated with poor outcome.^{5,6,8} Second, the model was based on patients from Europe in whom acute inflammatory demyelinating polyradiculoneuropathy is by far the most frequent manifestation of GBS. Prognosis in patients with GBS from other geographical areas, especially in those where the axonal variants are the predominant forms, might depend on other factors. Third, the Erasmus GBS outcome score might only be directly applicable to patients treated in hospitals in which high-level supportive and therapeutic care are available.

Nonetheless, the Erasmus GBS outcome score has the capacity to assist physicians to inform individual patients with GBS and their relatives about the prognosis of the disease and to guide decisionmaking with respect to treatment. Adequate prognostic models are also crucial to compare the results of previous therapeutic studies and to design future trials that could focus on more aggressive forms of treatment in patients at high risk for poor outcome. Previous studies showed that, despite treatment with plasma exchange and intravenous immunoglobulin, about 20% of patients will not regain the capacity to walk independently and about 5% still die. Moreover, additional forms of treatment are needed for GBS, especially for those patients who have a poor recovery after the established forms of treatment.^{2,24,25} Such patients in our model have an Erasmus GBS outcome score higher than 4. Future trials could focus on new forms of treatment for these poor prognostic subgroups, especially if such treatments are potentially hazardous and patients with a high chance of recovery need to be excluded.

Contributors

RvK participated in data collection, data analysis, and model development. EWS participated in data analysis and model development. RACH, AVS, and PAvD participated in data collection and data analysis. BCJ had the idea for the study and participated in data analysis and model development. All authors contributed to the writing of the manuscript and read and approved the final manuscript.

Conflict of interest statement

We declare that we have no conflict of interest.

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